

Supporting Information for

Growth Hormone Regulates the Stem Cell Population in the Growth Plate.

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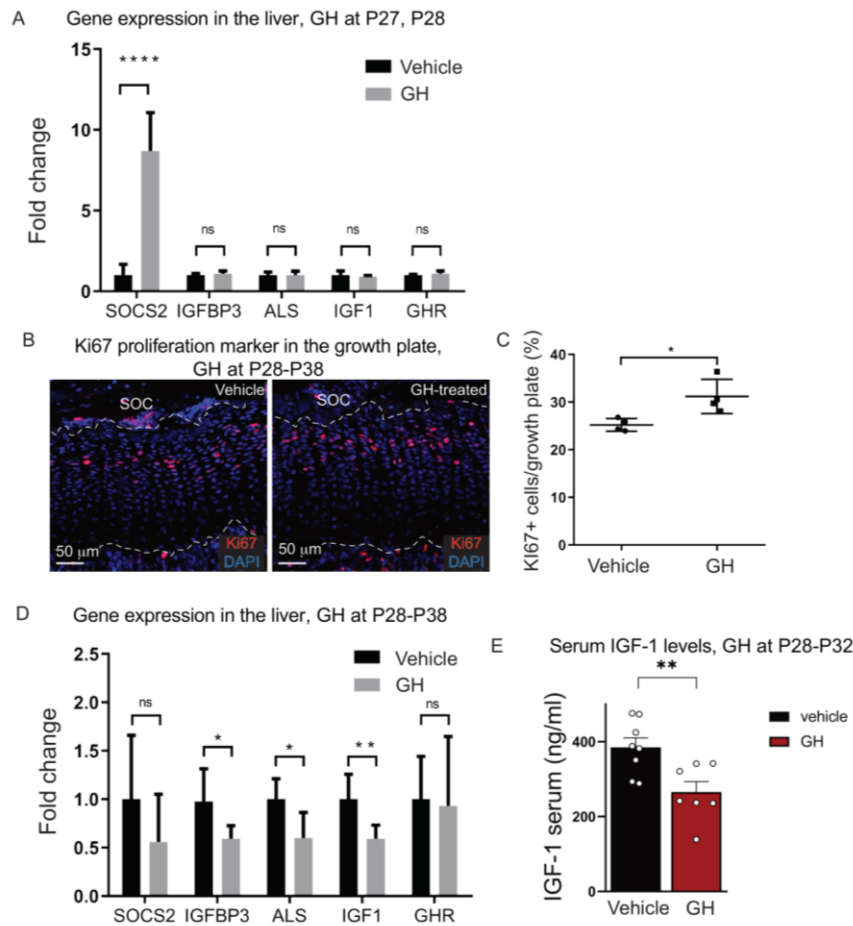


Fig. S1. GH modulates liver gene expression and increases proliferation in the growth plate.

plate.

(A) Quantitative RT-PCR of liver genes *Socs2*, *Igfbp3*, *Als*, *Igf1*, and *Ghr* after two daily GH injections.

(B) Representative images of Ki67+ proliferating cells (red) in tibial growth plates from vehicle and GH-treated mice. DAPI (blue) stains nuclei; white dashed lines mark growth plate boundaries. The secondary ossification center (SOC) is labeled for orientation.

(C) Quantification of Ki67+ cells. Mean $\pm$ SD, n = 4 mice/group. *p < 0.05 (unpaired t-test).

(D) Quantitative RT-PCR of liver genes *Socs2*, *Igfbp3*, *Als*, *Igf1*, and *Ghr* after ten daily GH injections.

(E) Measures of serum IGF-1 levels after four daily GH injections. Mean $\pm$ SE, n = 8 mice for control and n = 7 mice for GH-treated group. *p < 0.01 (unpaired t-test).

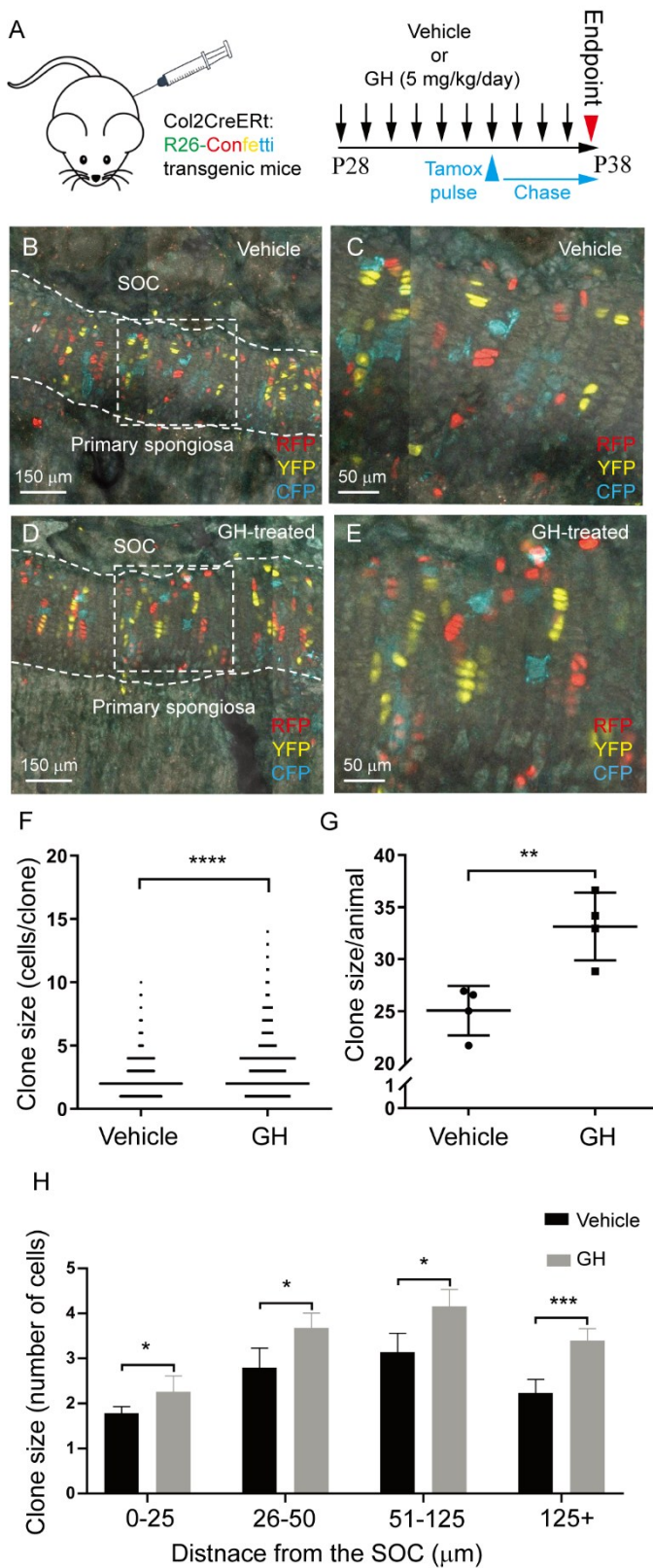


Fig. S2. GH increases proliferative activity across growth plate zones.

(A) Experimental design. Col2-CreERT2:R26R-Confetti mice were treated daily with GH or vehicle from P28 to P38 and received tamoxifen on P34.

(B–E) Representative confocal images of clones generated during the 4-day tracing period in tibial growth plates of vehicle- (B, C) and GH-treated (D, E) mice. Panels C and E show higher magnification of B and D, respectively. (F) Quantification of all individual clones in Col2-CreERT2:R26R-Confetti mice (n=4 mice/group) treated as in (A).

(G) Average clone size per animal.

(H) Quantification of clonal size stratified by distance from the secondary ossification center (SOC). Data represents mean $\pm$ SD, n = 4 mice/group. *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001 calculated by Mann–Whitney test (F), Unpaired t-test (G), and One-way ANOVA with Tukey's post hoc test (H). Confocal image panels were obtained by stitching adjacent tile scans of high magnification.

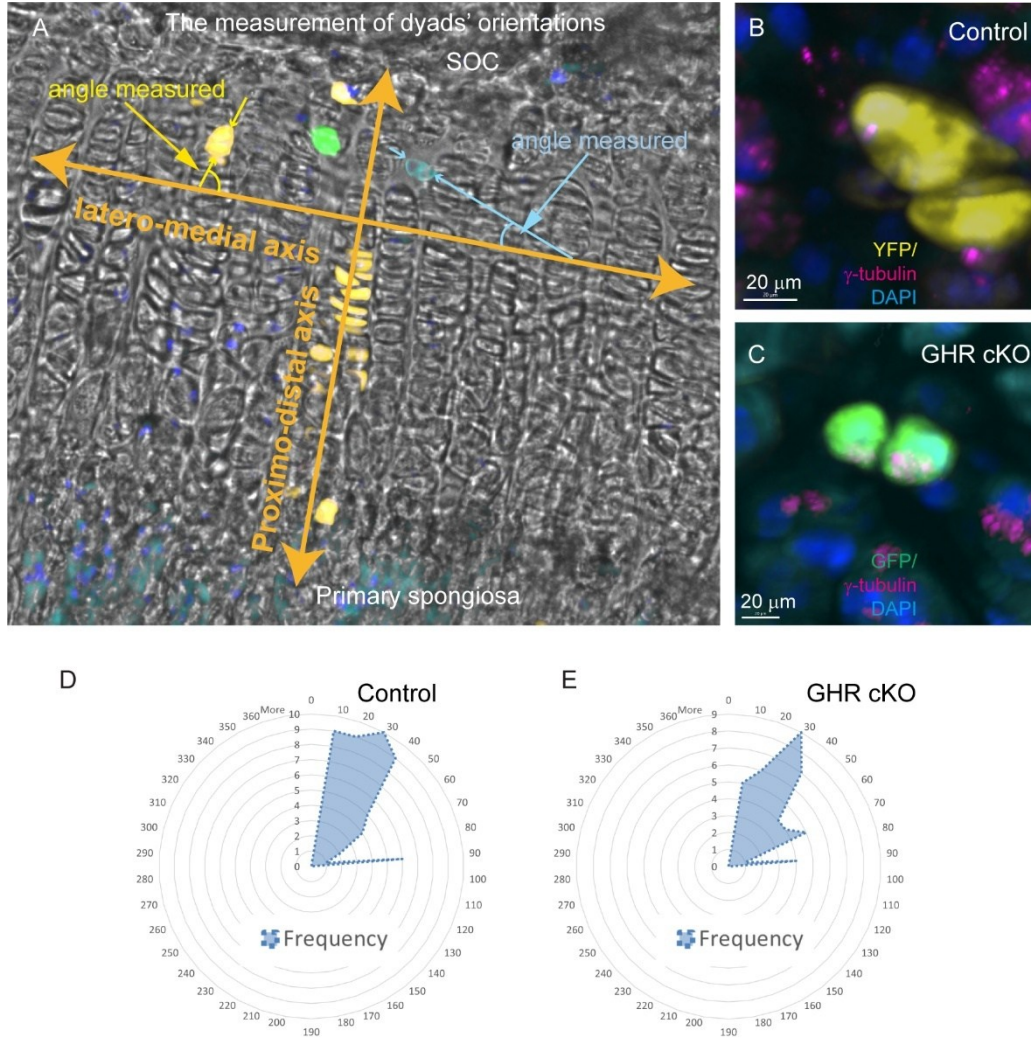


Fig. S3. Ablation of the growth hormone receptor in chondrocytes does not affect dyad orientation.

(A) Schematic of the approach used to quantify the orientation of clonal dyads in mice described in Figure 2A.

(B, C) Representative images of dyads from control (B) and GHR cKO (C) mice. Sections were counterstained with DAPI and γ -tubulin to enable accurate quantification.

(D, E) Quantification of dyad orientation in control (D) and GHR cKO (E) mice. SOC, secondary ossification center.

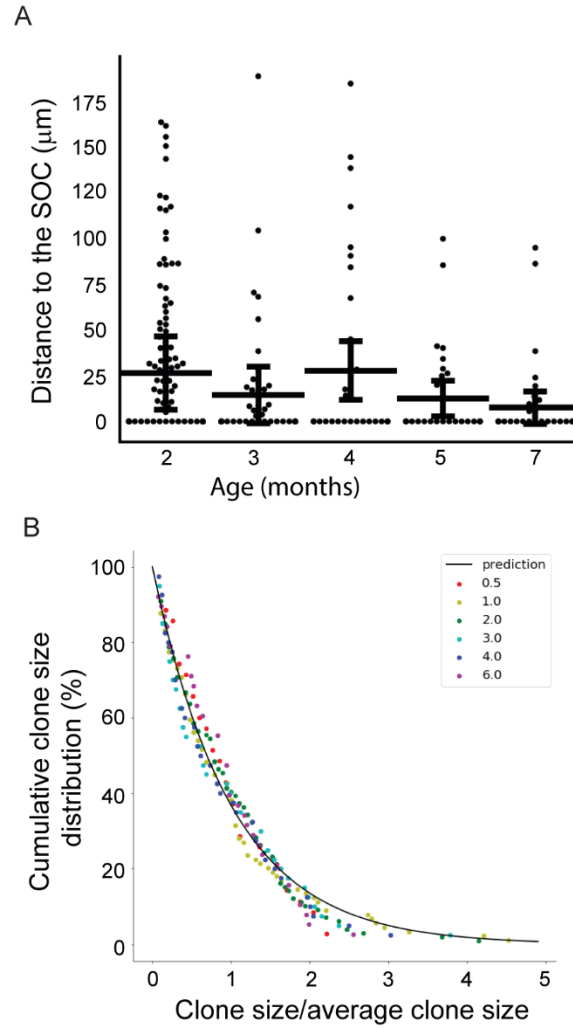


Fig. S4. Additional clonal dynamics in the growth plate.

(A) Distance from SOC to top of clones measured at 1–6 months post-tamoxifen in Col2 CreERT2:R26R-Confetti mice pulsed with tamoxifen at 1 month of age.

(B) Clonal scaling in Col2-CreERT2:R26R-Confetti mice pulsed with tamoxifen at P6, shown as clone size relative to the average at each time point.

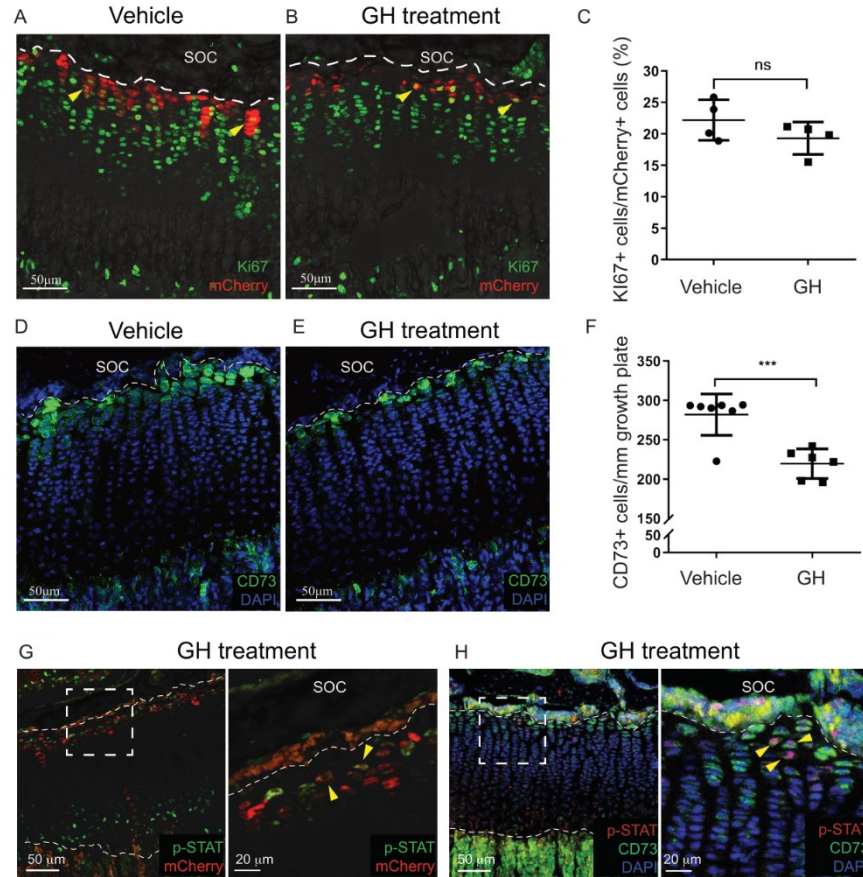


Fig. S5. Additional analysis of GH effects on stem cells.

(A–C) Confocal images showing Ki67 and mCherry co-staining in PTHrP-mCherry mice (vehicle vs. GH), with quantification (C).

(D–F) CD73 stem cell marker staining in tibial growth plates of vehicle- (D) and GH-treated (E) C57Bl/6 mice. Quantification shown in (F). n = 7 (vehicle) and 6 (GH); ***p < 0.001 (unpaired t-test).

(G, H) p-STAT5 co-localization with mCherry (G) or CD73 (H) marks the activation of GH signaling in stem cells (yellow arrowheads). Confocal image panels were obtained by stitching adjacent tile scans of high magnification.

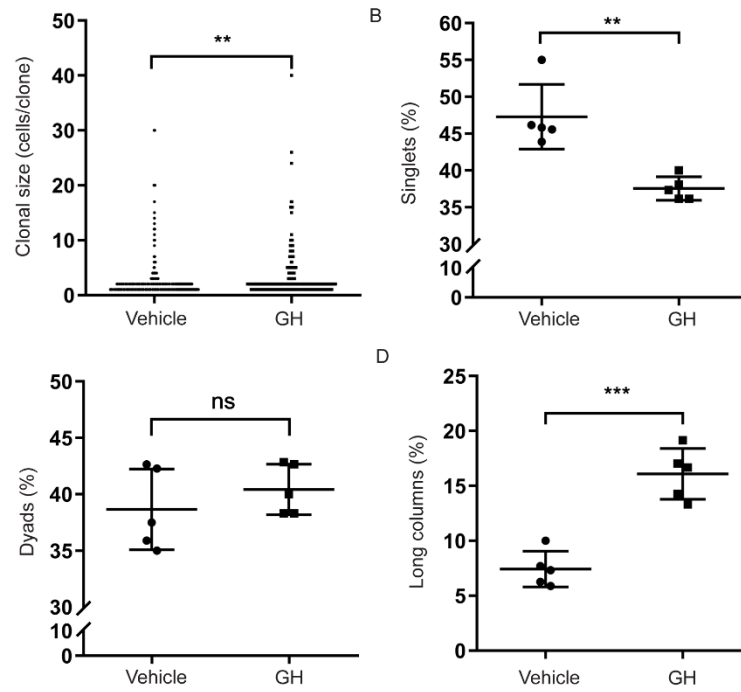


Fig. S6. GH regulates PTHrP+ stem cells in the femoral growth plate.

(A) Clone size quantification in femoral growth plates from GH- and vehicle-treated PTHrP-CreERT2:R26R-tdTomato mice treated as in Figure 6. ** $p < 0.01$ (Mann-Whitney test).

(B–D) Clone stratification into singlets (B), dyads (C), and large clones (D). $n = 5$ mice/group. ** $p < 0.01$, *** $p < 0.001$, ns – not significant (unpaired t-test).

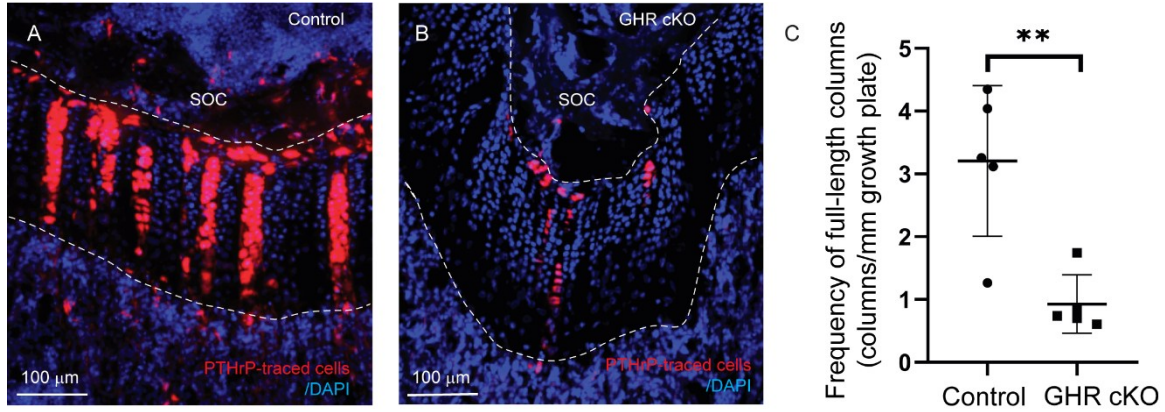


Fig. S7. Analysis of Femoral growth plates in GHR deficient mice.

(A-C) Femoral growth plates were analyzed in PTHrP-CreERT2:R26R-tdTomato mice presented in Figure 6: tamoxifen pulse at P6, analysis at P38.

(A, B) Representative confocal images of femoral growth plates showing chondrocyte column formation derived from PTHrP+ stem cells in mice with one (A, control) or two (B, GHR cKO) copies of GHR deleted. The growth plate is demarcated by dashed lines. The Secondary Ossification Center (SOC) is indicated for orientation.

(C) Quantification of Tomato-positive columns spanning the entire width of the femoral growth plate. A total of three 30 μm-thick sections were quantified and averaged for each mouse. Data are presented as mean ± SD, with n = 5 mice per group. **p < 0.01 (unpaired t-test).

Movie S1&2 (separate file). Three-dimensional light-sheet microscopy of cleared tibiae from 75-day-old control (Video S1) and GHR cKO PTHrP (Video S2) mice with tamoxifen administered at P6. Tomato-positive chondrocyte columns were labeled with anti-RFP antibodies, imaged throughout the entire tibia, and rendered for full-volume visualization and quantification.